

5.1.14 Factorial and "n Choose r" Notation

The product of all consecutive integers up to a given integer occurs so often in mathematics that it is given a special notation—*factorial* notation.

Definition

For each positive integer n , the quantity **n factorial** denoted **$n!$** , is defined to be the product of all the integers from 1 to n :

$$n! = n \times (n - 1) \times \dots \times 3 \times 2 \times 1$$

Zero factorial, denoted $0!$, is defined to be 1:

$$0! = 1$$